QUASI-COHERENCE OF SECTIONS WITH SUPPORT IN A RETROCOMPACT CLOSED SUBSET AND THE RIGHT ADJOINT PROPERTY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- For any object E there is a distinguished triangle relating sections with support, E itself, and the pushforward from the complement. invoked at proof lines 1053-1058. Provides the triangle used to prove quasi-coherence.
- For j the inclusion of a retrocompact open, the pushforward Rj_* preserves quasi-coherence of cohomology sheaves. invoked at proof lines 1059-1061. Gives that i_*R\mathcal{H}_T(E) has quasi-coherent cohomology sheaves, proving (1).
- The support functors form an adjoint pair between complexes on X and complexes with support in T. invoked at proof lines 1062-1064. Together with (1) identifies i_*R\mathcal{H}_T as the right adjoint on D_QCoh, proving (2).

Source reference. Original lemma 0G72: sources/stacks-chow/tags.txt, source lines 20217–20217. Cited at proof lines 1054-1055 for the triangle underlying both parts; the quasi-coherence and adjointness conclusions are inside the counted span.

Source reference. Original lemma 08D5: sources/stacks-perfect/pinned-original-source.tex, source lines 661–661. Cited at proof lines 1059-1060 to prove part (1); the conclusion of part (1) and part (2) are inside the counted span.

Source reference. Original lemma 0AEF: sources/stacks-chow/tags.txt, source lines 13123–13123. Cited at proof lines 1063-1064 for the adjointness behind part (2); the conclusion is inside the counted span.

Lemma 0.2. *Let X be a scheme. Let $T \subset X$ be a closed subset such that $X \setminus T$ is a retrocompact open of X. Let $i : T \to X$ be the inclusion.*

(1) *For E in $D_{QCoh}(\mathcal{O}_X)$ we have $i_* R\mathcal{H}_T(E)$ in $D_{QCoh,T}(\mathcal{O}_X)$.*
(2) *The functor $i_* \circ R\mathcal{H}_T : D_{QCoh}(\mathcal{O}_X) \to D_{QCoh,T}(\mathcal{O}_X)$ is right adjoint to the inclusion functor $D_{QCoh,T}(\mathcal{O}_X) \to D_{QCoh}(\mathcal{O}_X)$.*

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Proof. Set $U = X \setminus T$ and denote $j : U \to X$ the inclusion. By Cohomology, Lemma 0G72 there is a distinguished triangle

$$i_*R\mathcal{H}_T(E) \to E \to Rj_*(E|_U) \to i_*R\mathcal{H}_Z(E)[1]$$

in $D(\mathcal{O}_X)$. By Lemma 08D5 the complex $Rj_*(E|_U)$ has quasi-coherent cohomology sheaves (this is where we use that U is retrocompact in X). Thus we see that (1) is true. Part (2) follows from this and the adjointness of functors in Cohomology, Lemma 0AEF. $\square$